Excretion

Mark scheme and worked solutions for 15 questions: 5 multiple choice, 10 structured.

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1 · Part 5

Paper 1 & 2 Excretion [1]

Answer: C

Worked solution

The skin is one of the three organs of excretion in humans, along with the lungs and kidneys. Sweat contains water, ions and a small amount of urea. Of the options, only urea is a waste product of metabolism.

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2 · Part 5

Paper 1 & 2 Excretion [1]

Answer: B

Worked solution

Recall the photosynthesis equation: carbon dioxide + water → glucose + oxygen. The two products are glucose and oxygen. The plant uses the glucose, so the waste product is oxygen.

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3 · Part 5

Paper 2 only Excretion [1]

Answer: C

Worked solution

Urine flows kidney → ureter → bladder → urethra. The bladder is the only one of these structures that stores urine.

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4 · Part 5

Paper 2 only Excretion [1]

Answer: A

Worked solution

The order through a nephron is Bowman's capsule → proximal convoluted tubule → loop of Henle → second convoluted tubule → collecting duct. So the collecting duct comes last.

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5 · Part 5 Paper 2 only Excretion [1]

Answer: D

Worked solution

Ultrafiltration lets only small molecules into the nephron. Useful glucose is then reabsorbed. The fluid that remains, urine, contains water, urea and ions.

6 · Part 5 Paper 1 & 2 Excretion [5]

Mark scheme

(a) **[1]**

- skin (1)

(b) **[2]**

- total volume through organs of excretion = 1350 + 450 + 400 = 2200 cm^3 (1)
- (1350 ÷ 2200) × 100 = 61.4 % (1)

ALLOW 61.36 % OR 61 %

ECF correct use of the formula with an incorrect total scores 1 mark only (e.g. 1350 ÷ 2350 × 100 = 57.4 % scores 1)

(c) **[2]**

- excretion is the removal of waste products of metabolism OR of substances made by reactions in body cells (1)
- faeces are mainly undigested food that was never absorbed into the body OR was not made by body cells (1)

ALLOW removal of faeces is egestion (for the second mark only when linked to undigested food)

Worked solution

(a) Sweat is made by the skin, which is one of the three organs of excretion.

(b) The three organs of excretion are the kidneys (urine), the skin (sweat) and the lungs (exhaled air). Faeces are not produced by an organ of excretion, so leave them out. Total = 1350 + 450 + 400 = 2200 cm^3. Percentage = (1350 ÷ 2200) × 100 = 61.4 %. Check: 61.4 % of 2200 is about 1351 cm^3, which matches the urine volume.

(c) Excretion means getting rid of waste products made by metabolism inside cells, such as carbon dioxide and urea. Faeces are mostly food that was never digested or absorbed, so it never entered the body's cells. Removing faeces is called egestion.

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7 · Part 5

Paper 1 & 2 | Excretion | [5]

Mark scheme

(a) **[1]**

- respiration (1)

ALLOW aerobic respiration

(b) **[3]**

- carbon dioxide diffuses (1)
- from a higher concentration in the cell and air spaces to a lower concentration outside the leaf OR down a concentration gradient (1)
- (through the air spaces and) out through the stomata (1)

(c) **[1]**

- any one from: a leaf loses carbon dioxide by diffusion through the stomata but a human breathes it out from the lungs; a human carries carbon dioxide in the blood to an organ of excretion (the lungs) but a leaf loses it directly from its cells; a leaf has no ventilation (breathing) movements but a human does (1)

Worked solution

(a) Respiration in leaf cells releases carbon dioxide all the time. At night there is no photosynthesis to use it up, so the carbon dioxide is lost from the leaf.

(b) The carbon dioxide concentration in the cell is higher than in the air outside, so carbon dioxide diffuses down its concentration gradient. It moves out of the cell, into the air spaces in the leaf, and out through the stomata.

(c) Humans carry carbon dioxide in the blood to the lungs and breathe it out. A leaf has no lungs and no breathing movements; carbon dioxide simply diffuses out through the stomata.

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8 · Part 5

Paper 1 & 2 | Excretion | [7]

Mark scheme

(a) **[2]**

- 3.8 ÷ 0.04 (1)
- = 95 (times) (1)

(b) **[3]**

- carbon dioxide is produced by respiration in (body) cells (1)
- carbon dioxide is carried in the blood (plasma) to the lungs (1)
- carbon dioxide diffuses from the blood into the alveoli OR air in the lungs and is breathed out (1)

(c) **[2]**

- (muscle) cells respire faster OR rate of respiration increases (1)
- to release more energy (for muscle contraction), so more carbon dioxide is produced (1)

Worked solution

(a) 3.8 ÷ 0.04 = 95, so exhaled air has 95 times more carbon dioxide. Check: 0.04 × 95 = 3.8.

(b) Body cells respire and make carbon dioxide as a waste product. The blood plasma carries it to the lungs. The carbon dioxide concentration in the blood is higher than in the air in the alveoli, so carbon dioxide diffuses into the alveoli and is breathed out.

(c) Running muscles need more energy, so they respire faster. Faster respiration makes more carbon dioxide, so more is excreted by the lungs each minute.

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9 · Part 5 Paper 2 only Excretion [5]

Mark scheme

(a) **[4]**

- ball of blood capillaries: glomerulus (1)
- cup-shaped structure: Bowman's capsule (1)
- first coiled section: proximal convoluted tubule (1)

ALLOW (first) convoluted tubule

- U-shaped section: loop of Henle (1)

(b) **[1]**

- ureter (1)

REJECT urethra

Worked solution

(a) The nephron begins with the glomerulus, a ball of capillaries, which sits inside the cup-shaped Bowman's capsule. Filtrate then flows into the proximal convoluted tubule, the first coiled section, then around the U-shaped loop of Henle, then along the second convoluted tubule into the collecting duct.

(b) Each kidney is joined to the bladder by a ureter. The urethra is a different tube: it carries urine from the bladder out of the body.

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10 · Part 5

Paper 2 only

Excretion

[6]

Mark scheme

(a) **[3]**

- blood in the glomerulus is at high pressure (1)
- water and small molecules OR named small molecules (e.g. glucose, urea, ions) are forced out of the capillaries (1)
- into the Bowman's capsule (1)
- large molecules (protein) and blood cells stay in the blood (1)
- max 3

(b) **[2]**

- glucose, urea and (sodium) ions have the same concentration in the plasma and the filtrate (1)
- protein is present in the plasma but absent from the filtrate (1)

ALLOW quoted figures from the table

(c) **[1]**

- protein molecules are too large to pass through the capillary walls OR filter (into the Bowman's capsule) (1)

Worked solution

(a) Blood in the glomerulus is under high pressure. The pressure forces water and small dissolved molecules such as glucose, urea and ions out through the capillary walls into the Bowman's capsule. This liquid is the glomerular filtrate. This is ultrafiltration.

(b) The table shows glucose (1.0), urea (0.25) and sodium ions (3.2 g per dm^3) at the same concentration in both liquids, because these small molecules pass through freely. Protein is 76 g per dm^3 in plasma but 0 in the filtrate.

(c) Protein molecules are too big to pass through the filter, so they stay in the blood.

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11 · Part 5 Paper 2 only Excretion [8]

Mark scheme

(a) **[2]**

- glucose is needed (by cells) for respiration OR is a useful substance (1)
- otherwise it would be lost in the urine OR wasted (1)

(b) **[2]**

- glucose is reabsorbed by active transport OR against a concentration gradient (1)
- active transport needs energy, which is released by respiration in mitochondria (1)

ALLOW mitochondria release energy OR make ATP for reabsorption (second mark)

(c)(i) **[1]**

- 450 − 375 = 75 mg (1)

(c)(ii) **[2]**

- (375 ÷ 450) × 100 (1)
- = 83.3 % (1)

ALLOW 83 %

(d) **[1]**

- the cells can only reabsorb a limited mass of glucose each minute OR the reabsorption is already at its maximum rate OR too much glucose is filtered for all of it to be reabsorbed (1)

Worked solution

(a) Glucose is a useful substance that cells need for respiration. If it were not reabsorbed it would pass out in the urine and be wasted.

(b) All the glucose is removed from the filtrate, even when its concentration in the filtrate becomes lower than in the blood. That needs active transport, which uses energy. Mitochondria release energy by respiration, so cells with many mitochondria can carry out a lot of active transport.

(c)(i) Glucose lost = filtered − reabsorbed = 450 − 375 = 75 mg per minute.

(c)(ii) Percentage = (375 ÷ 450) × 100 = 83.3 %. Check: 83.3 % of 450 is about 375 mg.

(d) The cells can only reabsorb glucose up to a maximum rate. In untreated diabetes more glucose is filtered than this, so the extra glucose stays in the filtrate and ends up in the urine.

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12 · Part 5 Paper 2 only Excretion [7]

Mark scheme

(a) **[2]**

- ((9.0 − 1.5) ÷ 1.5) × 100 (1)
- = 500 % (1)

(b) **[4]**

- water is absorbed into the blood, so the water content of the blood rises (1)
- less ADH is released (by the pituitary gland) (1)
- the collecting duct (walls) become less permeable to water (1)
- less water is reabsorbed into the blood (1)
- so a larger volume of (more dilute) urine is produced (1)
- max 4

REJECT answers that say more ADH is released, for the second, third and fourth marks

(c) **[1]**

- the extra water had been removed, so the water content of the blood was back to normal and ADH release returned to normal (1)

ALLOW either part of the idea for the mark

Worked solution

(a) Change = 9.0 − 1.5 = 7.5 cm^3 per min. Percentage change = (7.5 ÷ 1.5) × 100 = 500 %. Check: a 500 % increase means the new value is 6 times the original, and 6 × 1.5 = 9.0.

(b) Drinking a lot of water raises the water content of the blood. The pituitary gland releases less ADH. With less ADH, the walls of the collecting ducts become less permeable to water, so less water is reabsorbed into the blood. More water stays in the urine, so the volume of urine goes up and it is more dilute.

(c) Once the extra water has been lost in urine, the water content of the blood is back to normal. ADH release goes back to normal and urine production returns to its starting rate.

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13 · Part 5

Paper 2 only

Excretion

[5]

Mark scheme

(a) **[3]**

- (water moves by) osmosis (1)
- from a dilute solution inside the collecting duct to a more concentrated solution in the surrounding fluid (1)
- through a partially permeable membrane OR the walls of the collecting duct (1)
- (and then) into the blood (capillaries) (1)
- max 3

(b) **[2]**

- more water moves out of the collecting duct OR more water is reabsorbed into the blood (1)
- so a small volume of very concentrated urine is produced, saving water OR less water is lost in urine (1)

Worked solution

(a) The fluid around the collecting duct is more concentrated than the fluid inside it. Water moves by osmosis from the dilute fluid in the duct, through the partially permeable walls, into the more concentrated surrounding fluid, and then into the blood.

(b) The more concentrated the surrounding fluid, the more water leaves the collecting duct by osmosis. So the kangaroo rat reabsorbs more water and makes a small volume of very concentrated urine. It loses very little water, which matters where there is almost nothing to drink.

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14 · Part 5 Paper 2 only Excretion [6]

Mark scheme

(a)(i) **[1]**

- the removal of waste products of metabolism (from the body) (1)

ALLOW named example, e.g. removal of urea made by the body

(a)(ii) **[1]**

- control (regulation) of the water content of the blood OR body (1)

ALLOW control of the balance of water and ions in the blood

(b) **[2]**

- 20.8 ÷ 0.32 (1)
- = 65 (times) (1)

(c) **[2]**

- (most of the) water in the filtrate is reabsorbed into the blood (1)
- much of the urea is not reabsorbed, so the urea is in a smaller volume of liquid OR becomes concentrated (1)

Worked solution

(a) Excretion means removing the waste products of metabolism, such as urea. Osmoregulation means controlling the water content of the blood.

(b) 20.8 ÷ 0.32 = 65, so urine has 65 times more urea than plasma. Check: 0.32 × 65 = 20.8.

(c) The kidney reabsorbs most of the water from the filtrate back into the blood, but much of the urea stays behind. The same urea ends up dissolved in much less water, so its concentration in urine is much higher.

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15 · Part 5 Paper 2 only Excretion [8]

Mark scheme

(a) **[6]**

- urea enters the glomerulus in the blood (1)
- blood in the glomerulus is at high pressure OR ultrafiltration occurs (1)
- urea is a small molecule, so it is forced out of the capillaries into the Bowman's capsule (1)
- urea becomes part of the glomerular filtrate (1)
- the filtrate passes through the proximal convoluted tubule, the loop of Henle and the second convoluted tubule (any two regions named in the correct order) (1)

ALLOW distal convoluted tubule for second convoluted tubule

- much of the urea is not reabsorbed (stays in the filtrate), while glucose is reabsorbed (1)
- urea enters the collecting duct, where water is reabsorbed, so the urea becomes more concentrated (1)
- urine passes down the ureter (1)
- urine is stored in the bladder (1)
- urine leaves the body through the urethra (1)
- max 6

(b) **[2]**

- glomerulus OR Bowman's capsule (the filter) (1)
- protein molecules are normally too large to be filtered, so a damaged filter lets protein through into the filtrate (and urine) (1)

ALLOW protein is not normally reabsorbed, so it must have entered the nephron through the filter (second mark)

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Worked solution

(a) Blood carrying urea enters the kidney and flows into a glomerulus. The blood there is at high pressure, so small molecules, including urea, are forced out of the capillaries into the Bowman's capsule. This is ultrafiltration, and the liquid formed is the glomerular filtrate. The filtrate flows through the proximal convoluted tubule, where glucose is reabsorbed but much of the urea stays in the filtrate. It then passes around the loop of Henle and along the second convoluted tubule into the collecting duct. In the collecting duct water is reabsorbed into the blood, so the urea becomes more concentrated. The fluid, now urine, flows down the ureter to the bladder, is stored there, and finally leaves the body through the urethra.

(b) Protein molecules are normally too big to pass from the glomerulus into the Bowman's capsule, so healthy urine has no protein. If protein appears in urine, the filter formed by the glomerulus and Bowman's capsule is probably damaged and is letting large molecules through.